

# Encoder and Decoder Structure

Deep Learning

Daniel E. Acuna

Associate Professor, University of Colorado Boulder

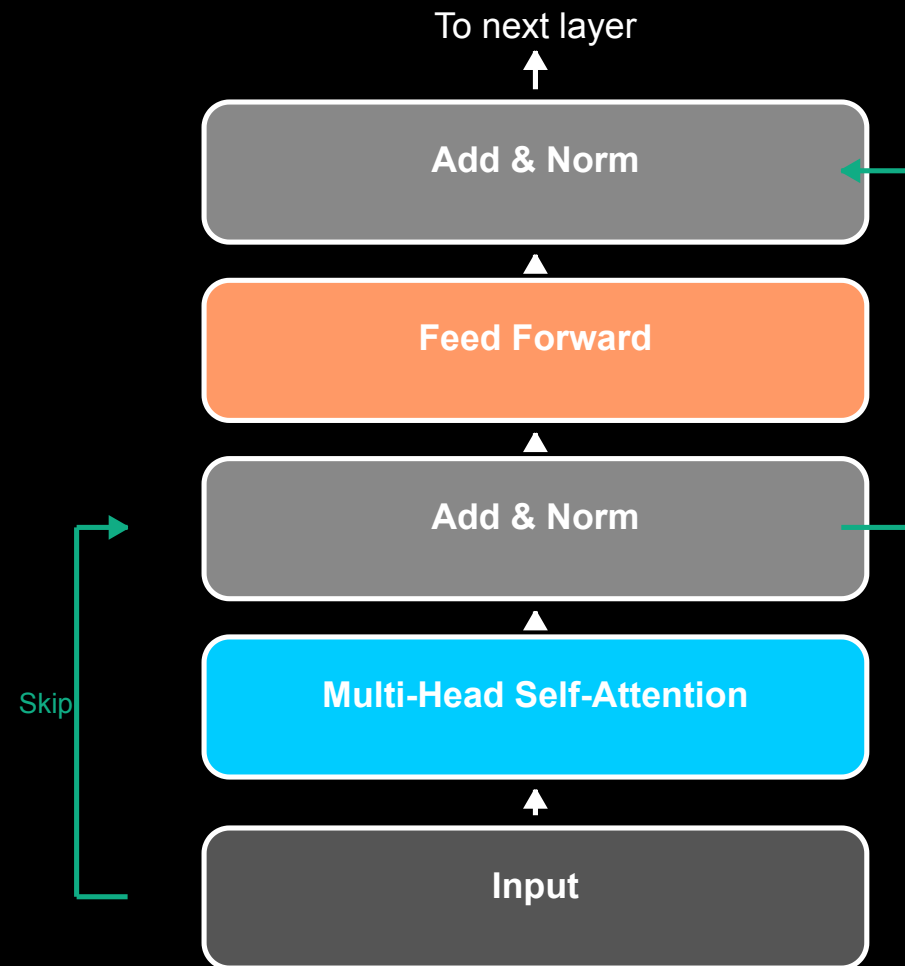
# Contents of This Video

In this video, we will cover:

- Encoder layer structure
- **Add & Norm: residual connections + layer normalization**
- **Positional encoding: how position is added**
- Decoder layer structure
- Masked self-attention in decoder
- Cross-attention: connecting encoder to decoder
- Information flow through the Transformer

# Encoder Layer

## Encoder Layer: Self-Attention + FFN



# Encoder: Bidirectional Self-Attention

Each position can attend to ALL other positions:

$$\text{Attention}(Q, K, V) = \text{softmax} \left( \frac{QK^\top}{\sqrt{d_k}} \right) V$$

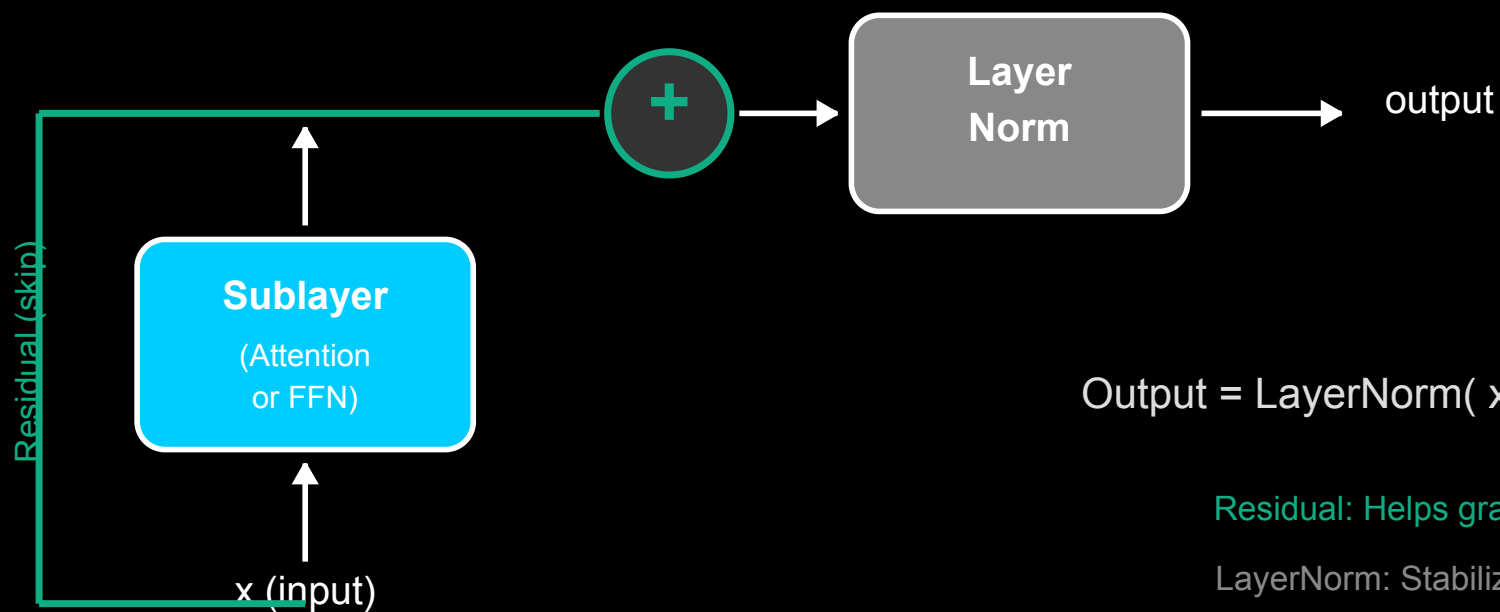
**“The cat sat on the mat”**

- “cat” can attend to “mat” (and vice versa)
- No masking—full visibility

**This is why encoders understand context so well!**

# Add & Norm: What's Inside?

Add & Norm = Residual Connection + Layer Normalization

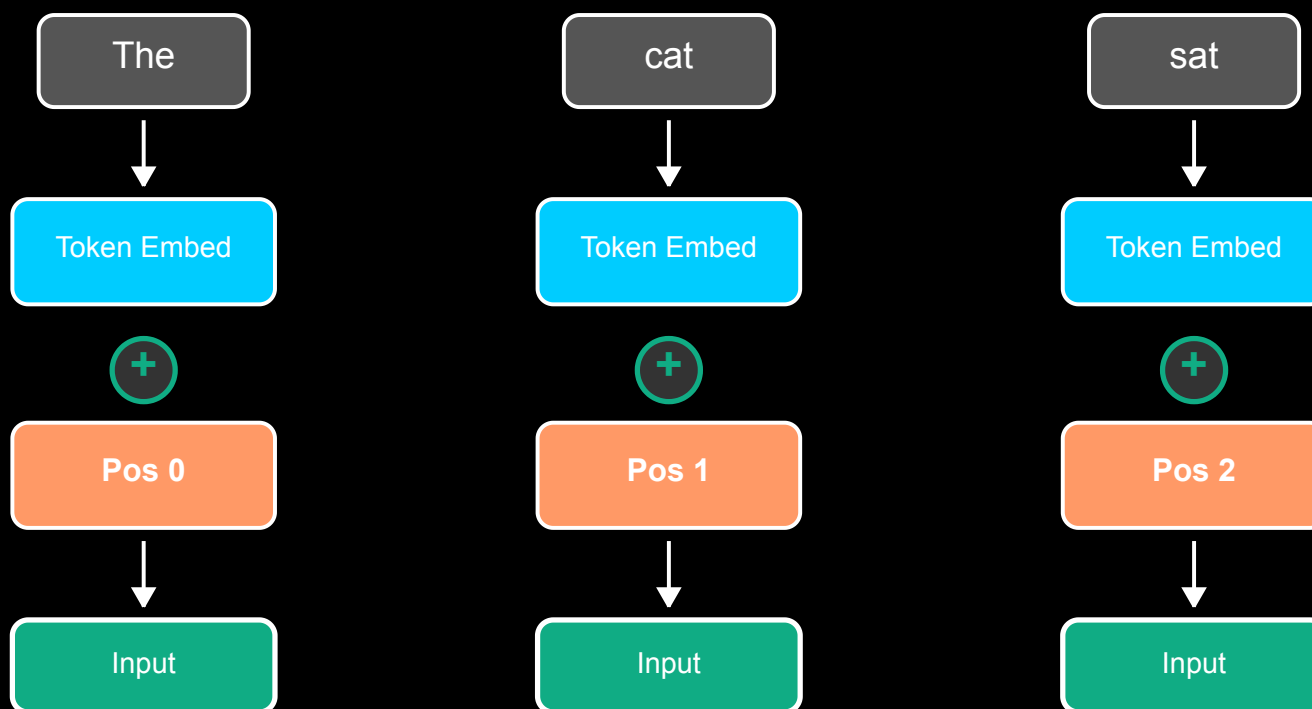


# Positional Encoding: Where Am I?

## Positional Encoding: Adding Position Information

Problem: Attention has no notion of order!

"cat sat" vs "sat cat" look the same to pure attention



### Solution:

Add position info  
to each token!

Sinusoidal (original):

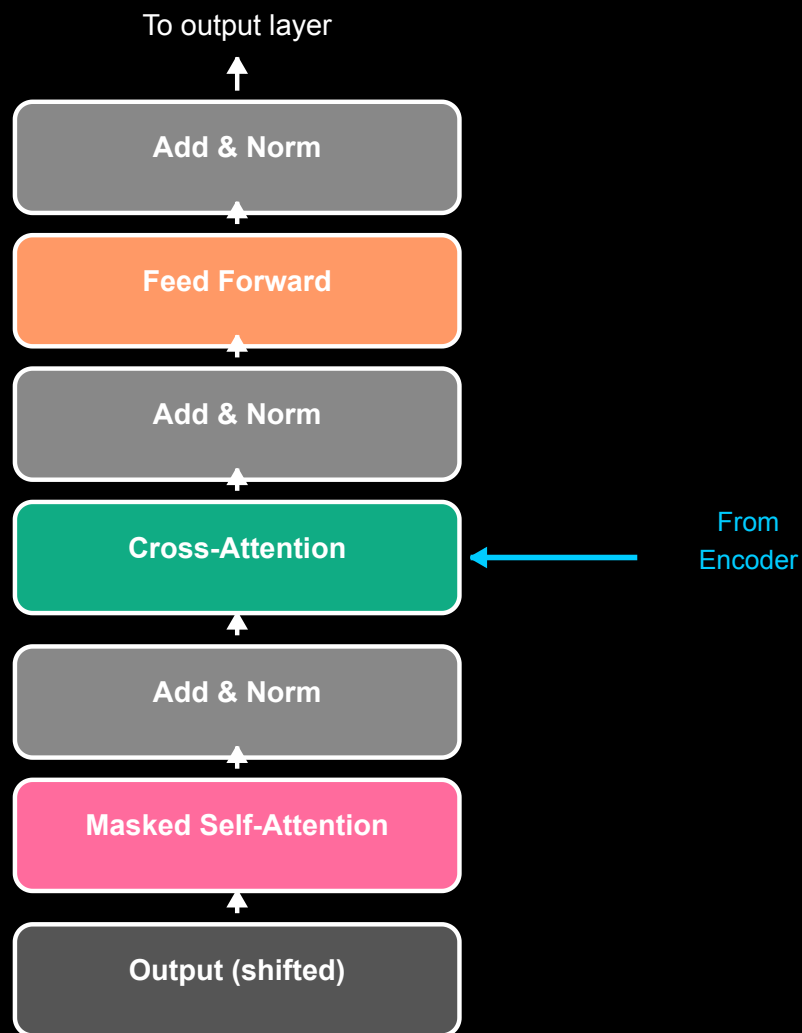
sin/cos functions

Learned (modern):

trainable vectors

# Decoder Layer

Decoder Layer: Masked Self-Attn + Cross-Attn + FFN

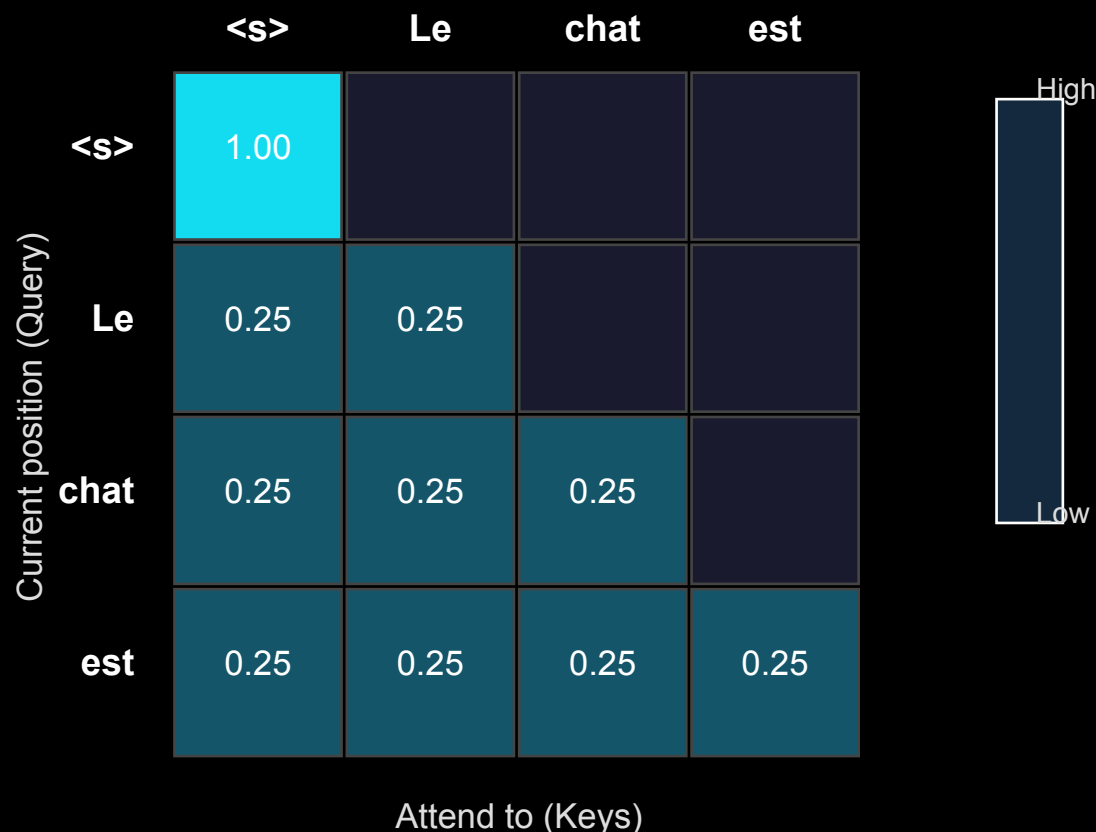


# Masked Self-Attention

**Problem:** During generation, can't see future tokens!

**Solution:** Mask future positions in attention

**Masked Attention: Can Only See Past**





# Cross-Attention

**Decoder attends to encoder output:**

$Q$  = decoder hidden states

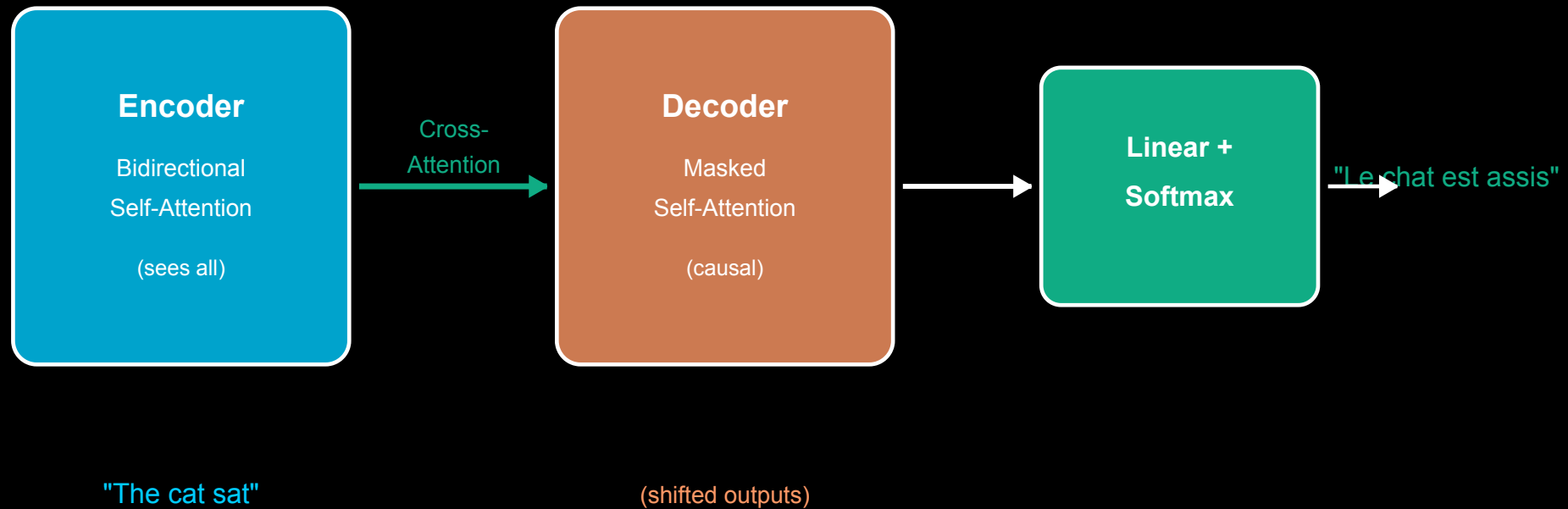
$K, V$  = encoder output

**What this does:**

- Decoder can “look at” the input when generating
- Different from self-attention:  $Q$  from decoder,  $K/V$  from encoder
- This is the bridge between input and output

# Full Information Flow

Transformer: Encoder Processes Input → Decoder Generates Output



# What We've Covered

In this video, we've learned:

- Encoder: bidirectional self-attention (sees all positions)
- **Add & Norm: residual connection + layer normalization**
- **Positional encoding: adds order information to embeddings**
- Decoder: masked self-attention (can't see future)
- Cross-attention: decoder queries encoder K/V
- Each layer: attention + FFN + residuals + layer norm